

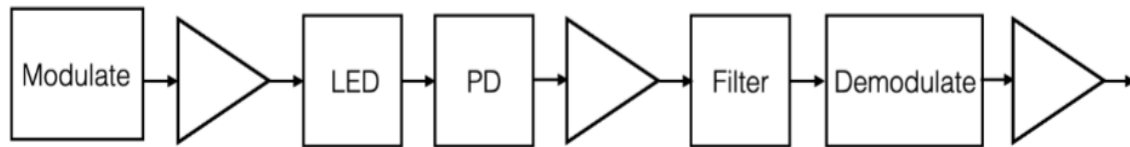
AM Radio

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June 12, 2024

1 Introduction

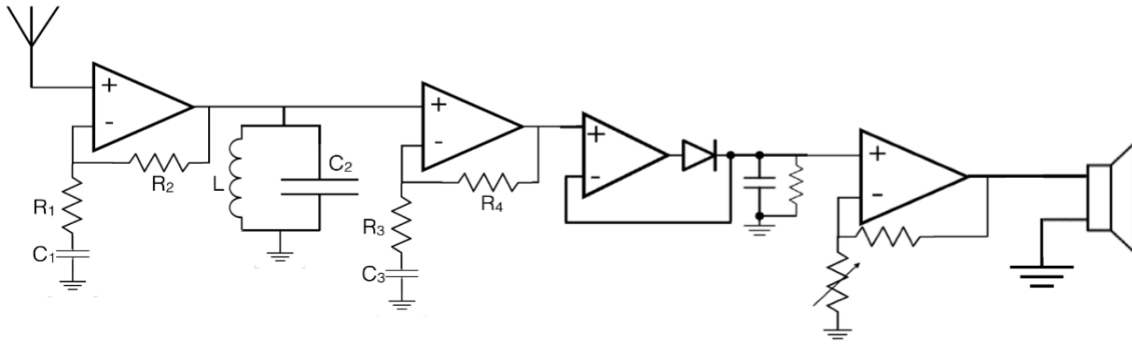
To design anything there must be very broad steps to take. In our radio we have two big circuit stages, one is responsible for generating the signal and the other is responsible for receiving the signal. Within these two stages there are specific areas such as the receiving stage needs a sensor for the signal, a way to process the information and a way to reveal that information, the delivery.



Let's look at how to generate the signal, we have to first take two signals which are the carrier signal and the message signal and then we'll somehow modulate(combine) both signals, there are different types of modulation but for the AM radio we'll use amplitude modulation. Note that our carrier frequency should be way bigger than our message frequency. The way we modulate this would be through a jFET which essentially multiplies both waves together and shows the modified amplitude. To generate the signal we want to pass it through an LED so it can communicate to a phototransistor. We should add an amplifier(I chose a noninverting amplifier) to make sure our modified signal can turn on our LED with no worry.

Now for the receiving stage(below) we want to detect the LED signal through a phototransistor and since our detected signal will be really small, we should amplify it with a non inverting amplifier. This should help the signal be more pronounced throughout the rest of the circuit. We then decided to add in the filter

which we chose to be a LC resonator circuit to make sure it captures the signal we want to see from the LED (match $\omega_0 = 1/\sqrt{LC}$ with signal). This circuit should also help out rule the frequency from the lights in the room. After the signal has gone through this stage it's time to demodulate it from the carrier, which we use a "leaky peak detector". In this demodulation we want to add an RC circuit (low-pass filter) with an RC constant equal to the message frequency. As for the last part of the circuit, we want to deliver the message by adding a noninverting amplifier that would serve as an audio amplifier which then is connected to a speaker.



2 Circuit Design

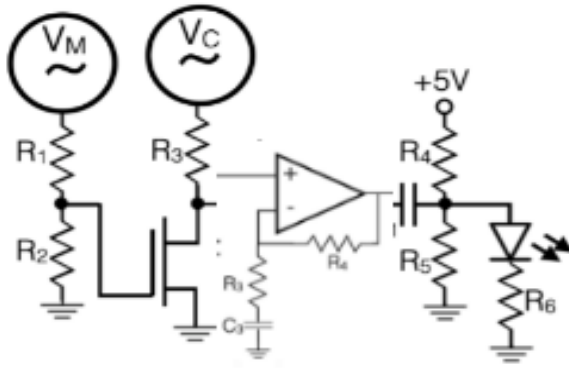
For all my noninverting amplifiers I used a resistor of 4.7k and 150, to produce large enough gain ($G = 1 + 4.7k/150$). All my op-amps are KA358, all my capacitors are $0.33\mu F$ except the ones used on the timer which were $4.7\mu F$ and $0.01\mu F$ as a bypass capacitor on the timer. For the JFET I used a J112 and the phototransistor used is a BPV11. The miscellaneous resistors like for the voltage dividers, I used 1.5k resistors.

As for my signals, my carrier frequency was 2kHz and my message frequency was 68Hz. So my $\omega_0 = 1/\sqrt{LC}$ has $L = 10mH$ and C is two $0.33\mu F$ in parallel to match message frequency. As for my peak detector I chose $R = 1.5k$ and $C = 0.33\mu F$

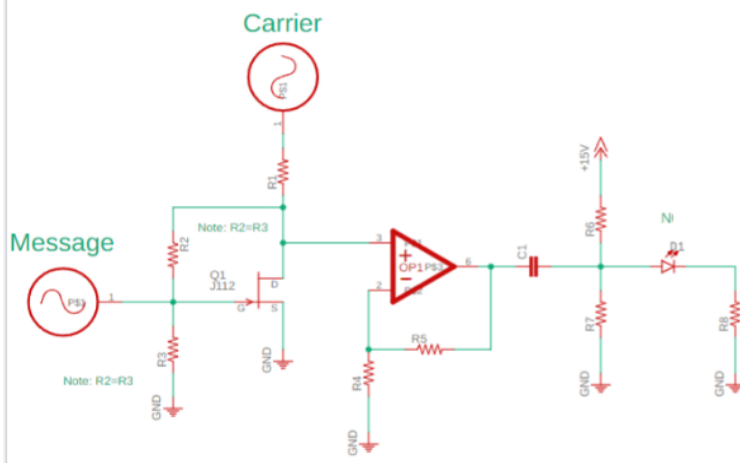
I did two circuits one with the timer in astable configuration and one with another sine wave. Both these message frequencies and amplitudes were the same, the only difference is that the square wave didn't have

negative voltages, it looked like pulses.

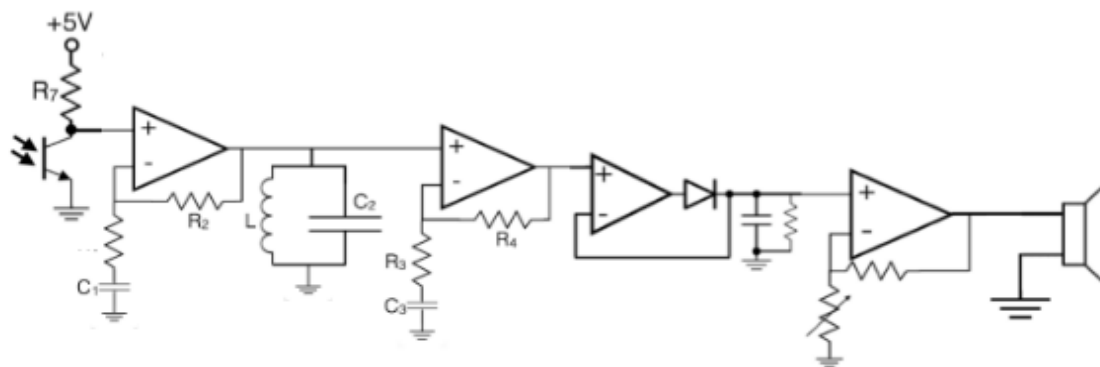
Below is the circuit for the signal Generator using a timer square wave as the message signal



Below is the circuit for two sine waves



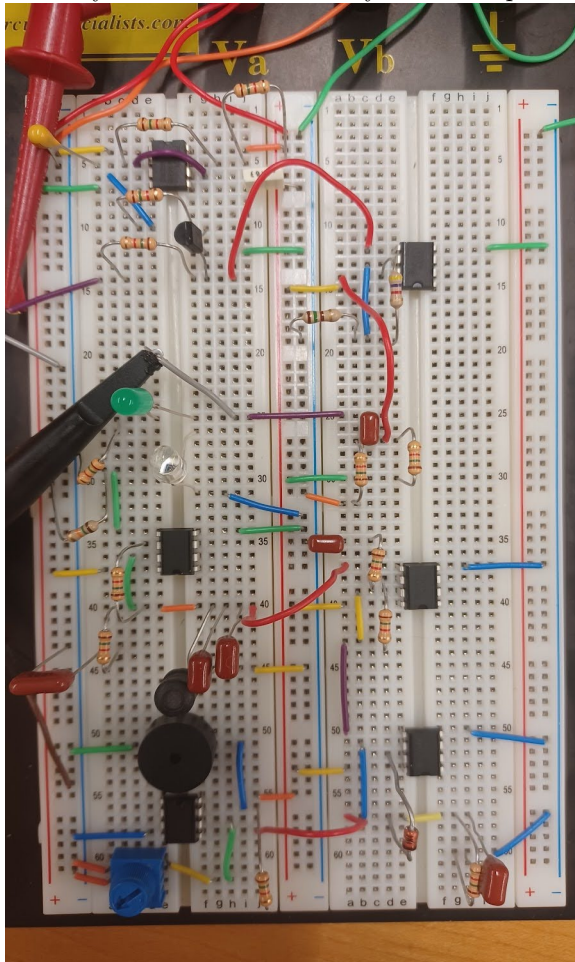
For both, I used this receiver circuit:



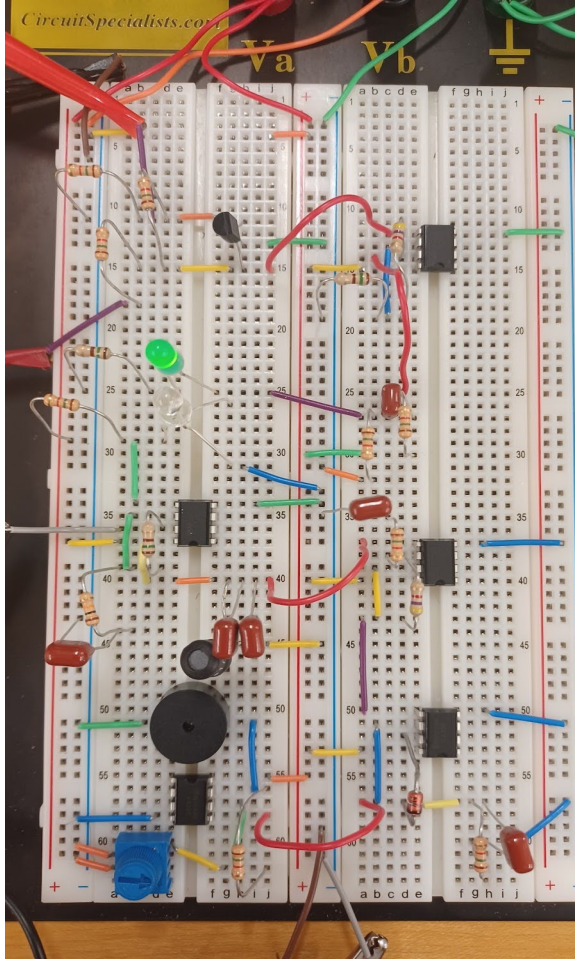
my 5V here was actually 10V.

3 Results

Below is my finalized circuit for my timer setup.



This below is my circuit for two sine waves.



Out of both circuits, my more successful one was the two sine wave circuit because I did more troubleshooting along the way. The LED turned on in this circuit, indicating I was doing something right. My signal was even detected by my phototransistor and was in a nice envelope when sent out. The modulation was good but not perfect. Modulation can't always be perfect in the real world, the modulated signal can have accumulated noise that can become more pronounced when amplified. The only thing I didn't get to work was the speaker.

4 Conclusion

This lab really helped me apply all the skills I learned including but not limited to troubleshooting, choosing resistor values, the importance of making filters and the use of voltage dividers. Before coming into this class, I thought electronics was really complicated and I thought it was extremely hard to know how people choose certain resistors, capacitors and other devices. Now I know that the reason for some of the capacitor and resistor values is to create a voltage divider or filter that'll have some rolloff frequency you want. All these little values can change the output of a stage and that can affect our entire circuit.

In conclusion, I think that the most fun skill I learned was troubleshooting, since now I can apply this skill on any electronic device. I would even say that troubleshooting includes knowledge of how to read circuits. Reading circuits another fun skill to have, but it can sometimes get complicated with many stages. Learning how to use the oscilloscope is another skill that I've built from this lab and it is one that can be used on any electronic device. I still think that electronics can get really complicated, so knowing these basics is essential if I continue down this route.

5 Lab Notebook

Google Doc URL [click here](#)